ModelIt!

Student Activity Pack

Inside the Cell: The Factory That Runs Your Body

How Mitochondria Power Every Move You Make

Grade 06L2

NGSS MS-LS1-1, MS-LS1-2

Energy and Matter: Flows, Cycles, and Conservation

STUDENT NAME _____ DATE _____

© 2026 Discovery Collective. All rights reserved.

Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

9 Post-Assessment 11

1 Pre-Assessment

ModelIt!

What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 Why do you think you need to eat food every day? What does food actually do inside your body?

2 Why do you need to breathe? What happens to the air once it enters your lungs?

3 What do you think provides energy to your cells? Where does that energy come from?

4 Draw what you think the inside of a cell looks like and label anything you know.

2 Scoring Rubric

ModelIt!

Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

ModelIt!

What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Nutrient Input | Oxygen Availability | Mitochondria Activity | Cell Growth Rate | Waste Production

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

ModelIt!

Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

ModelIt!

Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Healthy Cell

Moderate-high nutrients and oxygen

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Oxygen Deprivation

Lock Oxygen to 10%, nutrients normal

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Nutrient Starvation

Lock Nutrients to 10%, oxygen normal

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

ModelIt!

Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

ATP Concentration — The amount of energy currency available for cellular processes at any given moment

Temperature — Higher temperatures speed up chemical reactions in mitochondria, but extreme heat denatures enzymes

Cell Membrane Permeability — How easily nutrients and waste pass through the cell membrane, affecting supply and removal rates

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

ModelIt!

The Big Question

Why do you need to eat food AND breathe oxygen just to stay alive, and what happens inside your cells that makes both essential? Use evidence from your simulation to support your answers.

1. **Why do you breathe faster when you exercise? Connect this to what is happening inside your muscle cells.**

2. **What would happen to a cell if it had plenty of food but no oxygen?**

3. **How is a cell like a factory? What are the raw materials, the product, and the waste?**

4. **Why do some cells (like muscle cells) have MORE mitochondria than others?**

5. **How does the waste produced by your cells eventually leave your body?**

8 STEM Challenge: Design a Cell Survival System

ModelIt!

YOUR ENGINEERING MISSION

Design an emergency life-support system that keeps cells alive by maintaining optimal nutrient and oxygen delivery while removing waste products.

THE SCENARIO

A biotech company is designing an artificial organ support system for patients whose organs cannot supply cells with nutrients and oxygen efficiently. Your biomedical team must design a system that maintains cellular respiration in isolated tissue samples for 48 hours.

Engineering Design Thinking

1. HOW WILL YOUR SYSTEM DELIVER NUTRIENTS AND OXYGEN CONTINUOUSLY TO EVERY CELL IN THE TISSUE?

2. WHAT WASTE REMOVAL MECHANISM WILL PREVENT TOXIC BUILDUP OF CO_2 AND METABOLIC WASTE?

3. HOW WILL YOU ADJUST DELIVERY RATES BASED ON THE TISSUE'S METABOLIC DEMAND?

4. WHAT SENSORS WILL MONITOR ATP PRODUCTION, WASTE LEVELS, AND CELL HEALTH INDICATORS?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Cell Biologists and Biomedical Engineers study how cells function at the molecular level and design medical devices, artificial organs, and drug delivery systems. They earn $60,000-$130,000/year.

9 Post-Assessment

ModelIt!

Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 In the cell model, what happens when oxygen availability is locked at 10% while nutrients remain normal?

- **A.** The cell functions normally because nutrients alone provide enough energy
- **B.** Mitochondria activity drops dramatically because oxygen is required for efficient ATP production, causing cell growth to crash
- **C.** The cell switches to using a different organelle for energy production
- **D.** Waste production increases because the cell works harder without oxygen

2 Why does waste production increase when mitochondria activity increases?

- **A.** Mitochondria malfunction at high activity levels and produce extra waste
- **B.** Carbon dioxide and water are natural byproducts of cellular respiration, so more respiration produces more waste
- **C.** High mitochondria activity causes the cell membrane to leak, letting waste in from outside
- **D.** Waste production is random and not related to mitochondria activity

3 A student observes that oxygen deprivation kills cells in minutes, but nutrient starvation takes days or weeks. What does the cell model reveal about this difference?

- **A.** Cells do not actually need nutrients, only oxygen
- **B.** Oxygen is used faster than nutrients because cells breathe more than they eat
- **C.** Without oxygen, ATP production drops by 95% immediately. Without nutrients, cells can slowly burn stored reserves like glycogen and fat, declining more gradually.
- **D.** The difference is due to temperature, not the inputs themselves

4 How does the cell model explain why you breathe faster during exercise?

- **A.** Exercise heats up your lungs, making them expand and contract faster
- **B.** Muscle cells need more ATP during exercise, so mitochondria work harder, consuming more oxygen and producing more CO2 that must be exhaled
- **C.** Your brain signals faster breathing to cool your body down
- **D.** Exercise creates new mitochondria that require extra oxygen

5 Which relationship in the cell model creates a potential problem for cell health?

- **A.** Nutrient input positively affecting mitochondria activity
- **B.** Oxygen availability positively affecting mitochondria activity
- **C.** Waste production negatively affecting cell growth rate, because more energy production also creates more toxic byproducts
- **D.** Mitochondria activity positively affecting cell growth rate